ALETI VIHARI REDDY

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SUMMARY

Data Engineer with a Master's in Information Technology (3.86/4.0). Built an AWS event lakehouse processing **50M events** and a Kafka-Flink streaming pipeline processing **12.6M events**. Brings prior professional experience in business analysis, data validation and KPI definition.

EDUCATION

Belhaven University 01/2025 – 05/2026
M.S. Information Technology *Jackson, MS*

Jawaharlal Nehru Technological University 08/2016 – 09/2020
B.S. Computer Science *Hyderabad, India*

SKILLS

- **Languages:** Python, SQL, Java, Scala
- **Data Processing & Transformation:** Apache Spark, PySpark, dbt, Pandas
- **Streaming & Orchestration:** Apache Kafka, Apache Airflow
- **Data Platforms & Databases:** Databricks, Snowflake, PostgreSQL, MySQL, DuckDB
- **Cloud & DevOps:** AWS (S3, EMR), Docker, Git
- **Machine Learning & Gen AI:** Scikit-learn, LLMs, RAG, Prompt Engineering

SELECTED DATA ENGINEERING PROJECTS

AWS Event Analytics Lakehouse | 03/2026 – Present
Python, SQL, AWS, Airflow, Apache Iceberg

- Architected an **Airflow**-orchestrated S3 to Iceberg pipeline using Athena SQL and incremental merges into date-partitioned Parquet tables, processing 50M events in 25 minutes.
- Reduced one-day replay scan volume from 11.5 GB to 140 MB and runtime 30% by implementing event-date partitioning and partition-aware merges.
- Automated infrastructure and CI using **Terraform** and GitHub Actions, with 49 regression tests validating quality gates, atomic snapshot publication, and lock recovery.

Lakeflow Ingestion Framework | 09/2025 – 04/2026
Python, PySpark, Delta Lake, GitHub Actions

- Built a metadata-driven **PostgreSQL to Databricks** framework for full loads and WAL-based CDC for 2M+ records across 4 tables, preserving INSERT, UPDATE, and DELETE history
- Implemented three-attempt exponential backoff and classified transient versus non-recoverable failures, reducing unnecessary retries and improving recovery from transient errors.
- Automated multi-environment **CI/CD** with GitHub Actions and Databricks Asset Bundles, enforcing linting, type checking, and a 70% test-coverage gate.

Automated ETL Pipeline | 06/2025 – 10/2025
MySQL, Pandas, Scikit-learn, LLM, Docker

- Designed a **Bronze-Silver-Gold** ETL pipeline with Prefect, Polars, and DuckDB, integrating 4 source types (CSV, PostgreSQL, MySQL, S3) to produce curated analytics tables.
- Automated data-quality checks across 8 issue categories, using profiling, a weighted quality score, and rule-based remediation to produce auditable cleaned data.
- Built an **LLM-to-SQL** interface with schema-scoped context, query validation, and read-only database access to constrain generated analytics queries.

Real-Time Streaming Pipeline | 🔗 01/2025 – 04/2025

Python, Java, SQL, Kafka, Flink, Spark

- Engineered a Kafka-Flink streaming platform processing **12.6M events** in 30 minutes, decoupling KPI aggregation, anomaly detection, and downstream consumers.
- Developed 3 stateful anomaly detectors using Z-score, MAD, and EWMA to identify sudden metric spikes and gradual drift in streaming events.
- Designed analytics for a 5-stage conversion funnel by combining Flink per-event aggregation with **Spark** 30-minute session windows.

WORK EXPERIENCE

IT World Web 12/2022 – 12/2024

Executive Consultant – Business & Data *Bengaluru, India*

- Led UAT for migration of 2M+ customer records to a cloud warehouse, identifying 400+ defects pre-launch, zero critical issues in the first 90 days post-cutover
- Investigated **230 reporting** discrepancies in six months, traced 52% to inconsistent business definitions, and drove remediation that reduced repeat exceptions by 34%.
- Saved 12 hours per week in Weekly Business Review preparation by building **QuickSight dashboards** to analyze operational trends.

Careernet Technologies 04/2021 – 11/2022

Associate Consultant – Business & Data *Bengaluru, India*

- Standardized calculation logic for **10+ enterprise KPIs** by reconciling source data with performance reports, resolving conflicting metrics across stakeholder teams.
- Defined **source-to-target** mappings and data validation rules, reducing reporting defects by 35% and shortened analytics turnaround by one week per cycle.
- Mapped data collection steps in customer onboarding and defined requirements to reuse existing customer records, reducing manual entry time by 30%

CERTIFICATIONS & TRAININGS

- Analytical SQL for Developers – Oracle 07/2026
- Data Engineering on AWS Foundations – AWS 04/2026